

APPENDIX A. TRAINING AND IMPLEMENTATION DETAILS

For WTB wake supervision,

Training loop summary

Algorithm A1 Physics-aligned MoE training

Input: history windows X , graph sequence A , targets Y , primary regime anchors $\mathbf{p}_i$, auxiliary wake labels W (WTB only), model parameters θ

for each minibatch do

Build node-level graph context from the precomputed graph tensors

Encode X with the directed-diffusion GRU to obtain node context $\mathbf{u}_{i,t}$ Only MPPT and pitch-control samples

Compute gate logits Z and soft routing probabilities $\mathcal{A}_t(i,j)$ with defined wake flags. The graph-smoothness penalty is

Compute expert forecasts and aggregate them with G

Evaluate L_{pred}

If MoE mode is active:

add L_{bal} on top-K routing statistics

add L_{align} on valid primary regime anchors

if WTB: add L_{force} on MPPT/pitch samples

if WTB: add L_{aux} on wake labels

add L_{smooth} on the final graph snapshot

Update θ with AdamW and mixed precision

end for

Select the checkpoint with the best validation RMSE

$$\mathcal{L}_{\text{smooth}} = \frac{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i,j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i,j) + \epsilon}.$$

Graph construction details

For WTB, the downstream unit vector is $\mathbf{u}_{i,t} = [\sin(\theta_{i,t} + \pi), \cos(\theta_{i,t} + \pi)]^\top$. With $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$, the streamwise and cross-stream distances are $d_{ij,t}^\parallel = \Delta \mathbf{p}_{ij}^\top \mathbf{u}_{i,t}$ and $d_{ij,t}^\perp = |\Delta \mathbf{p}_{ij}^\perp \mathbf{u}_{i,t}|$. A candidate edge activates when

Model-family and supplementary evidence index

TABLE I
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Table A1. In-family model comparison used for mechanism validation.

Model	Routing	ExtraParams terms
Dense (matched)	Single dense head	None
Unconstrained MoE	Node-level soft gate	WTB 110,012 / ERA5 loss 103,272
Corrected routing comparator	Node-level soft gate	WTB 110,012 / ERA5 loss 103,843 on-ly forced: $L_{\text{bal}} +$ $L_{\text{align}} +$ L_{force} ; ER- A5: ful- l cor- rect- ed s- tack

$$\mathbb{I}_{ij,t}^{\text{cone}} = \mathbf{1} \left[d_{ij,t}^\parallel \geq 0 \wedge \arctan \left(\frac{d_{ij,t}^\perp}{\max(d_{ij,t}^\parallel, 10^{-6})} \right) \leq \phi \right].$$

Shared-mapping reference
The wake weight is $\tilde{\mathcal{A}}_t(i,j) = \exp(-d_{ij,t}^\parallel / \alpha) \exp(-|d_{ij,t}^\perp| / \beta) \mathbb{I}_{ij,t}^{\text{cone}}$. If wind direction is missing, fall back to $\mathcal{A}^{\text{static}}(i,j) = \exp(-\|\Delta \mathbf{p}_{ij}\|_2 / d_{\text{max}})$. Only the strongest M absolute weights are retained. The wake score and flag are $s_{i,t}^{\text{wake}} = \sum_j \mathcal{A}_t(i,j)$ and

$$W_{i,t} = \begin{cases} 1, & s_{i,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & s_{i,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

For ERA5, the retained symmetric graph uses a Gaussian kernel on great-circle distances:

$$\mathcal{A}(i,j) = \exp \left(-\frac{d_{ij}^2}{2\sigma^2} \right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

where σ is the median retained neighbor distance on the training graph.

Auxiliary losses

Let $\mathcal{B}$ denote the routed samples after masks, with $B = |\mathcal{B}|$. The soft importance and normalized share of expert e are $I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}$ and $P_e = I_e / \sum_r I_r$. The top-K load is $f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(g_n)]$, giving

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

Shared constants

TABLE II
***Table A3.** Shared architecture, graph, and training constants used in the reported experiments.

Item	WTB
History / horizon (H, P)	(36, 24)
Gate softmax temperature τ	0.7
Top- K in L_{bal}	1
Hidden size d	64
k_{nn} candidates	8
Maximum candidate radius d_{max}	1500 m
Wake-cone half-angle ϕ	25°
Streamwise decay α	1200 m
Cross-stream decay β	400 m
Retained inbound edges M	5
Optimizer	AdamW
Learning rate	2×10^{-3}

Dataset-specific thresholds and routing weights

ERA5
(36, 24)

Role

Input
Table III
fore*
cast**Table A4.** Dataset-specific region thresholds used to construct routing anchors.

Dataset	Soft routing sharpness	Threshold
WTB	Balancing s-tatic only Shared encoder width Initial neighborhood size	u_{idle}
WTB	WTB proximity	
WTB	ity fall-back / can-date filter	P_{th}
WTB	WTB downstream activation	$q_{0.75}^{\text{wake}}$
ERA5	WTB wake attention	$\varepsilon_{\text{sshf}}$
ERA5	WTB wake attention	$q_{0.95}^{\Delta}$
AdamW	Shared	
	op-ti-miz-er	
	Shared	

TABLE IV
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Table A5. Active routing-loss weights in the reported corrected models.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence